

XEVM-DM270GHK

TMS320DM270GHK Evaluation Module

Installation Guide Ver0.92 02/12/2003



TEXAS INSTRUMENTS JAPAN LTD.

DCES COMPANY

Imaging Products

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1. Preface

This Installation Guide is for operation of XEVM-DM270GHK board by users. It mainly describes power supply, jumpers, various switches, volume, and configuration and start-up of CCS.

This guide describes about **XEVM-DM270GHK Ver. 2.O**.

Refer to User's Guide that is available for the details of this board.

Please be sure to fill in DSC Development Tool User Registration Form included in this board and send it by facsimile to our facsimile number described on that form for sending User's Guide and information about this board.

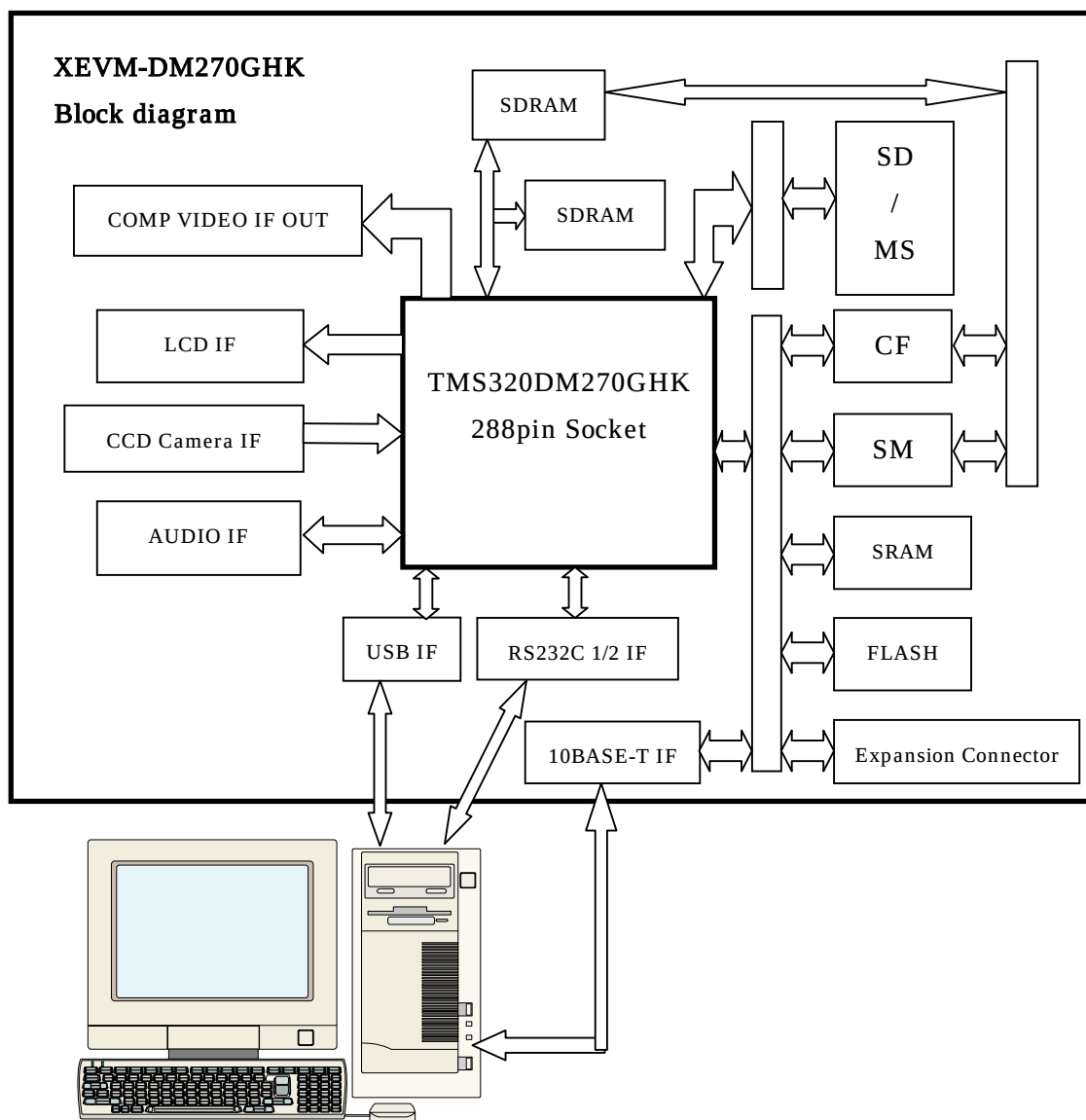
In addition, various information may be sent by e-mail. Fill also your e-mail address in DSC Development Tool User Registration Form.

1-1. Outline of XEVM-DM270GHK Evaluation Board

XEVM-DM270GHK is the board for evaluation of the device “TMS320DM270GHK” which is dedicated for Digital Still Camera applications.

This board includes CCD camera interface, video output of composite, four types of memory cards, communication ports such as RS232C and 10BASE-T, memories such as SRAM, FLASHROM, SDRAM, JTAG Emulation terminal for connecting with XDS510 Emulator, Audio I / O such as MIC/LINE input and LINE / Headphone output, terminals for the logic analyzer connection and so forth.

Block Diagram



1-2. Requirements for operation of TMS320DM270GHK

- External power supply

DC power supply at minimum +5V 2A

Connect the DC cable that comes with TMS320DM270GHK board. (Notice: Confirm the polarity)

- XDS510 Emulator (JTAG cable+ISA bus) or XDS510 PP, etc.

- CCS debugger

- Ver. 2.00 for ARM TMS470

- Ver. 2.00 for DSP TMS320C54x

Install the above debuggers on PC in advance.

- This TMS320DM270GHK evaluation board

- +(Plus) or -(minus) driver for voltage adjustment

When you use USB interface, RS232C interface or AUDIO interface, prepare the cables below separately.

RS232C cable

Cross connecting cable of 9-pin DSUB female type which is called "inter-link cable".

Cables connected with J5, J6, J7 and J8

Use cables for general-purpose stereo mini plug.

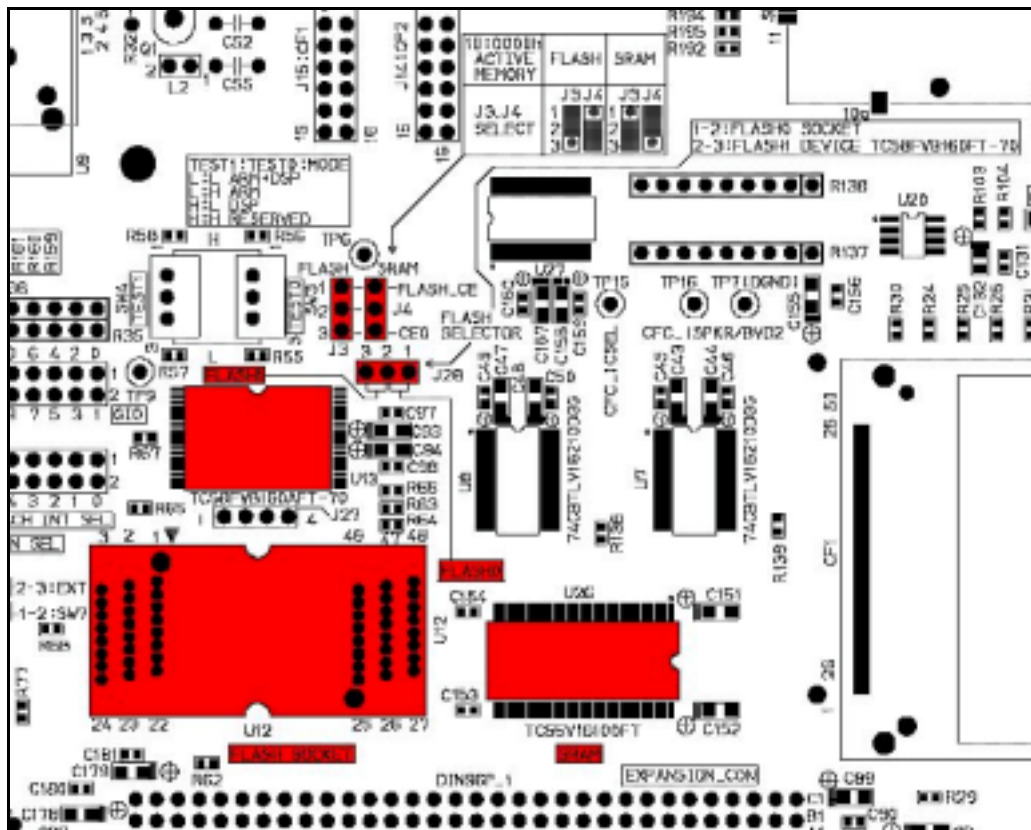
The connector of J8 microphone input is stereo mini plug jack, but this input is monaural. Therefore, you can use a cable for monaural mini plug for this terminal. In case of connecting stereo cable, L channel signal is connected with monaural input for this microphone.

2. Setting Confirmation of each Jumper

Before the application of power, confirm the setting of jumper below.

Memory switching jumper J3, J4, J28

Memory switching jumper J3, J4 and J28 are allocated on circuit board as follows.



J3 and J4 decide if FLASH_CE or CE0 will be used as chip select for SRAM or FLASH. Furthermore, it is possible to select a mounted FLASH (FLASH 1) or FLASH (FLASH 0) of a socket by J28.

On the memory from 0x100000 address selected by FLASH_CE, when

- SRAM is used,
-> J3: 2-3 SHORT, J4: 1-2 SHORT
- FLASHROM is used,
-> J3: 1-2 SHORT, J4: 2-3 SHORT
- Switching FLASH
-> J28 1-2 SHORT: Select mounted FLASH U13 (FLASH 1)
-> J28 2-3 SHORT: Select FLASH U12 (FLASH 0) of a socket

(Notice: Do not set up J3 and J4 jumpers on same position)

For example, in case that J3 is 1-2, J4 is 1-2. By placing them on same position, collision of both memory outputs is caused and it may damage memory on board.

- FLASHROM available

TC55V1610FT 16M bit FLASHROM manufactured by Toshiba Corp.

2-1. Default jumper setting

Left side of circuit board

JP16, 17, 18: 1-2 SHORT

JP1: ALL SHORT

JP22, 23: SHORT

JP2, 19: OPEN

When you use U1 TLV320AIC23PW, set jumpers as above.

GIO21 and 22 are used for AIC initialization.

JP4, JP5: 1-2 SHORT

These are jumpers for power supply for CCD camera. Use them by setting above.

SW1: ALL OFF

CCD PIN (19, 17, 15, 13, 11, 9, 7) PULL DOWN SWITCH (ON: GND)

Upper central of circuit board

JP6, JP11, JP7, JP12, JP13, JP14, JP8, JP9, JP10: ALL SHORT

These are jumpers for electric current measuring and all are shorted except during measuring of consumption electric current. (Insert jumper pin before use. Device doesn't operate normally without connecting. Only these jumper plugs have contact points outside so that you can measure their voltage by tester or voltmeter.

Center bottom of circuit board

JP24, 25: OPEN

TEST3(JP24), TEST4(JP25) are pulled up by SHORT.

SW2: ALL ON

1:EXTHOST, 2:BTSEL, 3:FLASHSEL and 4:TEST2 are pulled down by "ON".

SW6: ALL OFF

1:GIO8, 2: GIO9, 3: GIO10 and 4: GIO11 are pulled down by "ON".

JP26: 2-3 SHORT

1-2 SHORT: PWDN switch SW7 is used for the control of PWDWN terminal.

2-3 SHORT: PWDWN terminal signal is connected with external terminal drawn in DIN96P_1. PWDN SW SW7 is not used.

J1: 1-2 SHORT

J2: 2-3 SHORT

At the time of use of a DSC25 compatible CCD camera module, J1and J2 are made a setup as shown above so as to switch GIO21 and GIO27of the connector JE1. At this time, GIO21 signal is outputted in the pin 28 (GIO27PIN) of JE1 and GIO27 signal is outputted in the pin 22 (GIO21PIN).

When a DSC25 compatible camera isn't used, use J1: 2-3 SHORT, J2: 1-2 SHORT.

At this time, Pin 28 (GIO27PIN) is GIO27 and Pin 22 (GIO21PIN) is GIO21.

J25: 1-2 SHORT

MCBSP0 CLK Select

1-2 SHORT: OSC/EXT

2-3 SHORT: CLKOUT2 (GIO18)

JP21: ALL OPEN

ATTACH INT Select

1-2 SHORT: use GIO0 for ATTACH INT

:

:

15-16 SHORT: use GIO7 for ATTACH INT

Upper right of circuit board

JP30, JP32: SHORT

JP34, JP35: OPEN

Connecting TXDx and RXDx with 9-pin connector.

JP30: RXD0, JP32: TXD0, JP34: RXD1, JP35: TXD1

Right middle of circuit board

JP27: OPEN

Connecting an interrupt from Network IC with EXT_INT4 (GIO4).

SHORT: Connecting INTRQ0 of CS8900A-CQ3 with EXT_INT4 (GIO4) of DM270.

OPEN: INTRQ0 of CS8900A-CQ3 is N.C.

JP28: OPEN

Connecting AEN,SBHE of Network IC with CE1 of DM270 by short.

SHORT: Connecting AEN,SBHE of CS8900A-CQ3 with CE1 of DM270.

OPEN: AEN,SBHE of CS8900A-CQ3 are N.C.& Pullup

JP29: 1-2 SHORT

This selects the power of SDRAM.

1-2 SHORT: D3.3V (Digital 3.3V)

2-3 SHORT: D3.3VSD (controlled by VR6 D3-3VSD ADJ)

Bottom right of circuit board

J3: 2-3 SHORT

J4: 1-2 SHORT

28: 1-2 SHORT

Chip selection of 10_0000h Active memory is done.

SRAM is selected by default.

FLASH0 (socket) is selected as Flash Memory by default.

J3: 1-2 SHORT -> FLASHROM IS SELECTED BY FLASH_CE

J3: 2-3 SHORT -> FLASHROM IS SELECTED BY CE0

J4: 1-2 SHORT -> SRAM IS SELECTED BY FLASH_CE

J4: 2-3 SHORT -> SRAM IS SELECTED BY CE0

J28: 1-2 SHORT -> FLASH0 (socket) IS SELECTED

J28: 2-3 SHORT -> FLASH1 DEVICE (TC58FVB160FT-70) IS SELECTED

2-2. Voltage setting of EVM adjustable power supply

1. Connect a power cable with DC power supply. Do not apply the power yet.

2. Open GHK socket U9 and take TMS320DM270GHK out.

3. Apply the power by turning SW8 down to ON side.

4. Set up the power supply voltage by rotating LSI1.5V VR4 and LSI3.3V VR5.

Setting at shipping inspection: LSI1.5V = 1.5V, LSI3.3V = 3.3V

- Voltage setting of LSI1.5V

Put the tester on TP2 (DGND) and metal part of JP5 D1.5VINT and rotate LSI1.5V VR4, and then set the voltage as 1.5V.

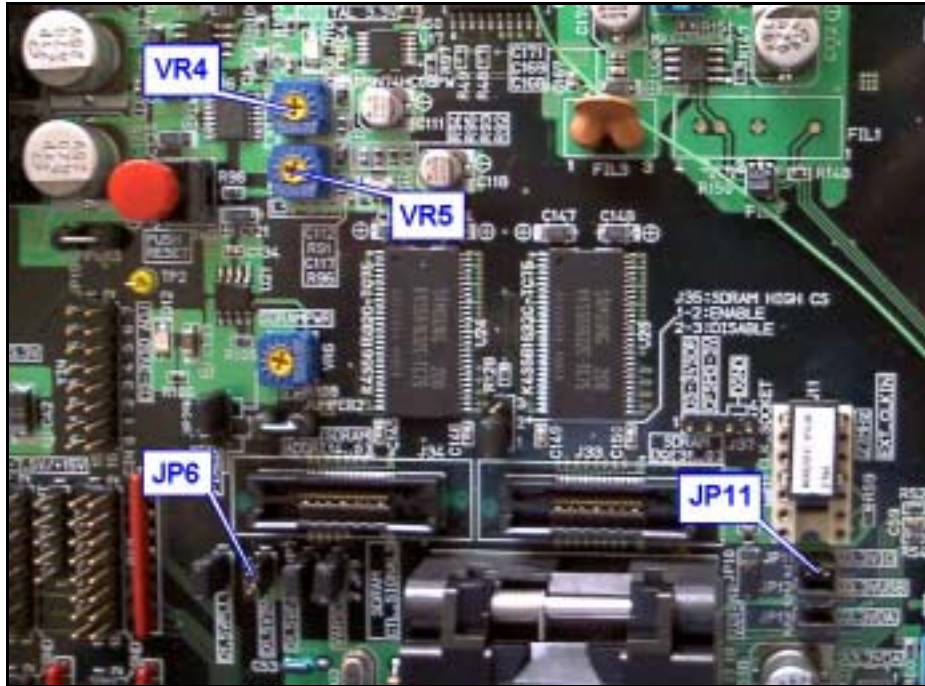
- Voltage setting of LSI3.3V

Put the tester on TP2 (DGND) and metal part of JP11 D3.3VIO and rotate LSI3.3V VR5, and then set the voltage as 3.3V.

5. After setting completed, turn SW8 down to OFF side and power off.

6. Open the cap of U9, put TMS320DM270GHK back.

Setting of power supply voltage for DM270 device was completed.



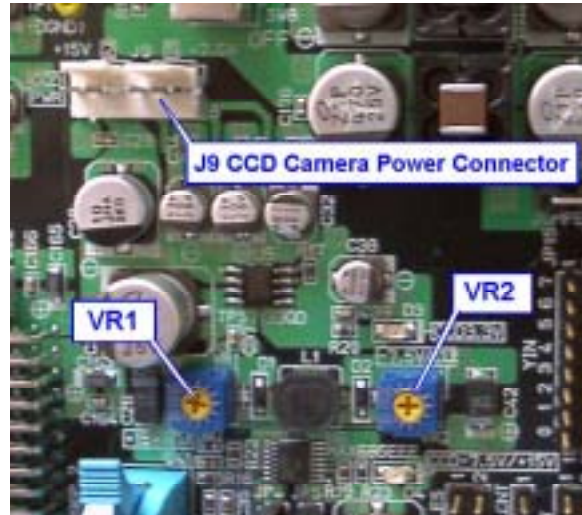
2-3. Voltage setting of CCD power supply

Setting of $+15V$

Rotate VR1 and measure the voltage between TP11 DGND and Pin 2 of CCD camera power supply connector J9 by tester, and then set it as +15V.

Setting of $-7.5V$

Rotate VR2 and measure the voltage between TP11 DGND and Pin 5 of CCD camera power supply connector J9 by tester, and then set it as -7.5V.



2-4. Adjustment of each COMPOSITE video output

Adjust each COMPOSITE video output by rotating each VR described below.

CAMPGAIN

VR7: Adjustment of signal level of COMPOSITE signal video

VREF RGB

VR for reference voltage adjustment of video DAC is VR3.

- Adjustment of VR in case of COMPOSITE video signal

Connect the monitor display to J36 COMPOSITE video output terminal and adjust VR below by watching monitor so that adequate image is displayed.

CAMPGAIN VR7: Adjustment of signal level of COMPOSITE signal video

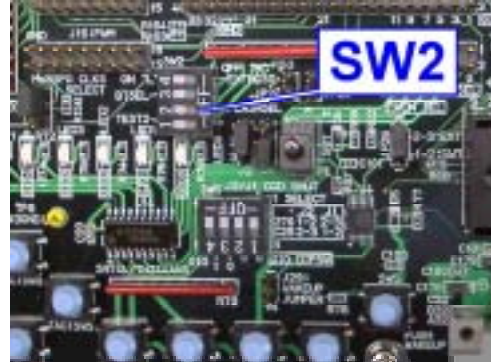
VREF RGB VR3: Reference voltage adjustment of video DAC



2-5. Setting of S2 that sets up DM270 operation mode

By using this S2, you can change the setting below.

It is ALL ON (LOW) by default.



HOST mode switching EXTHOST(4 of S2)

EXTHOST(4 of S2)	TMS320DM270GHK operation mode
ON "L" level	Internal ARM processor mode
OFF "H level	External host mode

ARM BOOT selector switching BTSEL(3 of S2)

BTSEL(3 of S2)	TMS320DM270GHK operation mode
ON "L" level	Boot from external FLASH memory.(SELF BOOT UP)
OFF "H level	Boot from ARM built-in memory.(Serial/Host BOOT UP)

FLASH ROM BUS selector switching FLSHSEL(2 of S2)

FLSHSEL(2 of S2)	TMS320DM270GHK operation mode
ON "L" level	SDRAM uses SDRAM bus. FLASHROM uses external memory bus.
OFF "H level	SDRAM and FLASHROM both use SDRAM bus.

ARM clock input source switching TEST2(1 of S2)

TEST2(1 of S2)	Clock used
ON "L" level	MXI (Q2 27MHz crystal resonator)
OFF "H level	EXT_CLK (Module mounted on U11 CLK SOCKET) 27MHzOSCmodule

By rotating each bit switch of S2 to

OFF side: It is OFF, "1" pull-up connection

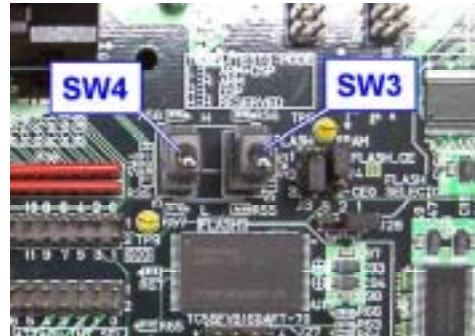
Another side: It is ON, connecting with "0" GND

2-6. Setting of emulation mode switch, SW3 and SW4

Determine the mode in case of debugging of DM270 with CCS (Code Composer Studio).

SW4 is **TEST1** and SW3 is **TEST0**.

Signal level of both terminals is set as “L” level by turning each SW down to front side (direction to bottom of circuit board) and “H” level by turning to far side (direction to top of circuit board).



The table below shows SW4 **TEST1**, SW3 **TEST0** and their corresponding emulation device on CCS.

SW4 TEST1	SW3 TEST0	Corresponding Emulation Device
L Front side	L Front side	ARM+DSP
L Front side	H Far side	ARM only
H Far side	L Front side	DSP only
H Far side	H Far side	Not available

(Notice: In case of rotating both switch, SW3 and SW4 to far side, emulation of device is not conducted normally.)

As for these SW3 and SW4, their **setting at emulation should not be changed on the way after starting-up of CCS**. Or else, CCS does not operate normally because of inconsistency with emulation mode set on CCS.

2-7. Caution on usage of XDS510

Please note the items below when you connect XDS510 emulator with this EVM board.

- **Insert and remove a XDS510 cable with the power status of EVM board OFF.**

In other words,

Do not insert / remove a XDS510 cable when EVM board is ON.

- When EVM board is OFF, remove the XDS510 cable from the EVM board.

In case, the XDS510 cable is kept plugged to the EVM, power off the PC connected with XDS510.

When you finish work, especially, do not keep XDS510 cable plugged into EVM board with leaving PC on for many hours. When PC is on, XDS510 cable is energized.

Even when EMV board is OFF, conducting XDS510 cable drives EVM board. In this case, XDS510 cable is damaged and may be broken.

3. Starting-up of this board and CCS

We describe the starting-up method here, assuming that CCS ARM ver. 2.00 and DSP ver. 2.00 are already installed.

3-1. Connecting the EVM board

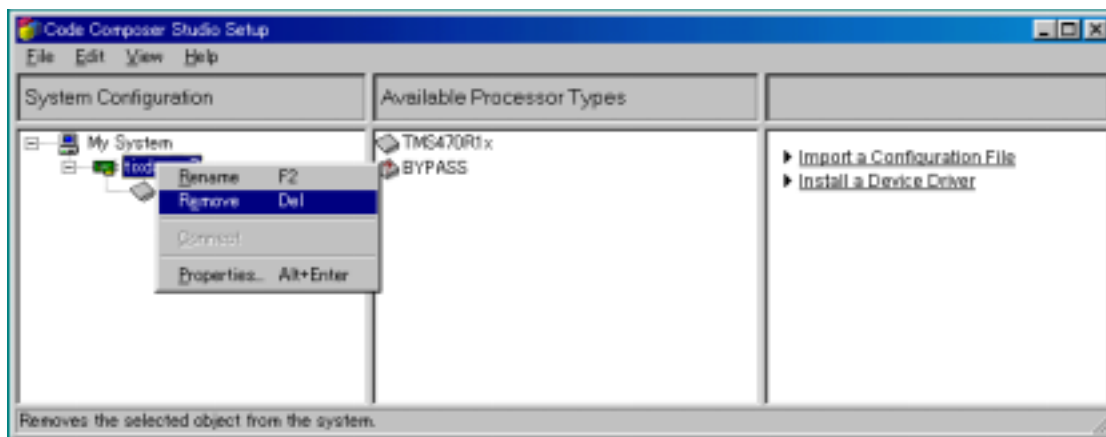
Connect DC 5V 2A-power supply, power cable and XDS 510 cable with EVM board. Do not apply DC 5V 2A-power supply yet.

Execute a set-up program to start up the CCS in heterogeneous (ARM+DSP) debugging mode.

PC: Start -> Program -> Texas Instruments -> Code Composer Studio 2 (OMAP) -> Setup Code Composer Studio

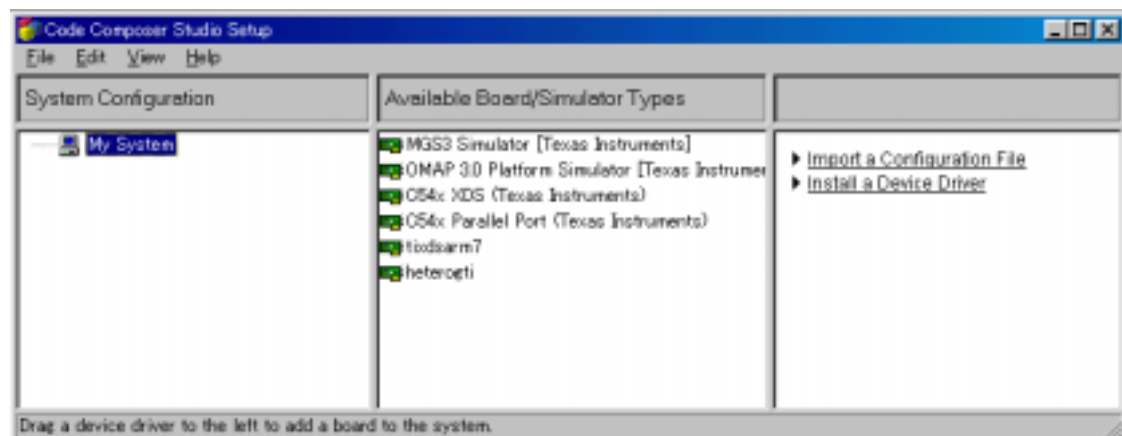
An image below appears. (ARM7 XDS is set for example in figure below)

Move a cursor to tixdsarm7 and click right button. You are asked to remove a board or not.



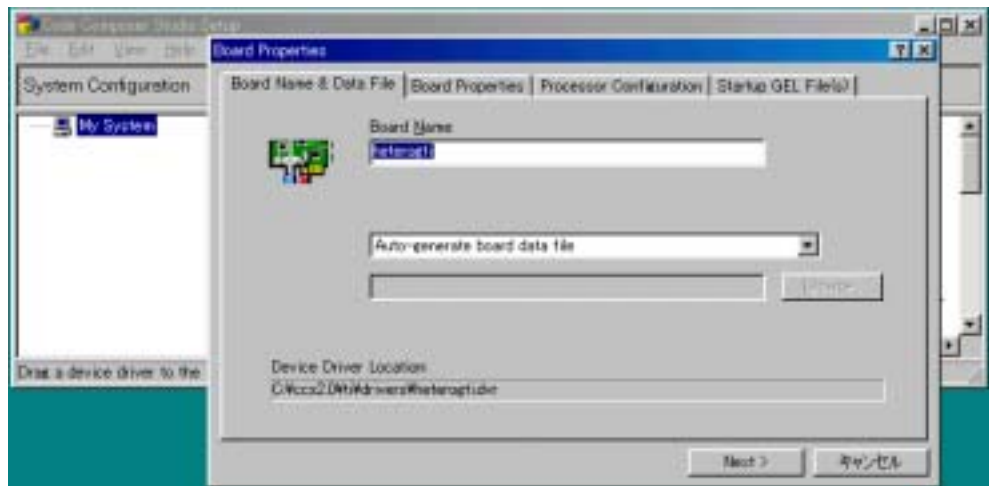
Move a cursor to "Remove Del" and click left button. Then you are asked again, so click the Yes (Y) button.

After pressed Yes (Y) button, it will be in non-setting status as below.

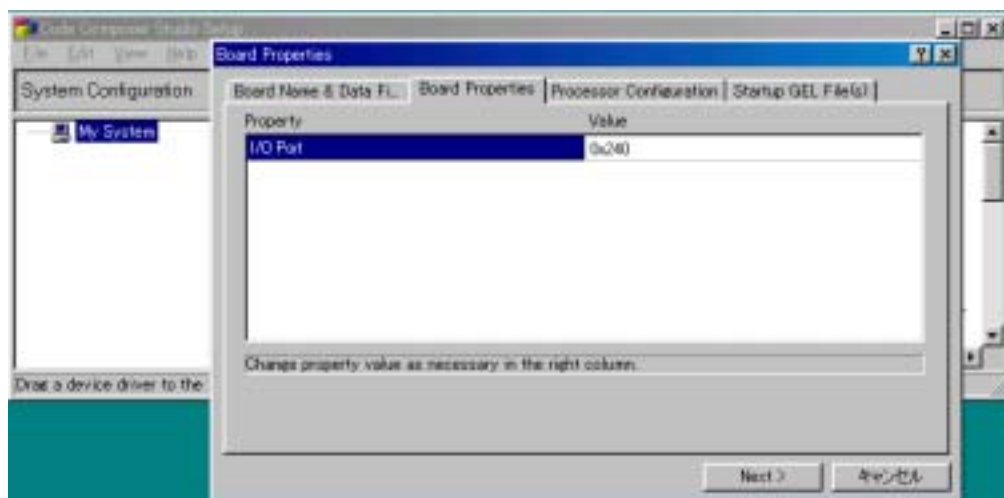


Drag “Heterogti” from “Available Board/Simulator Types” from center box to “System Configuration” in the left.

And then, a pop-up window opens as below.

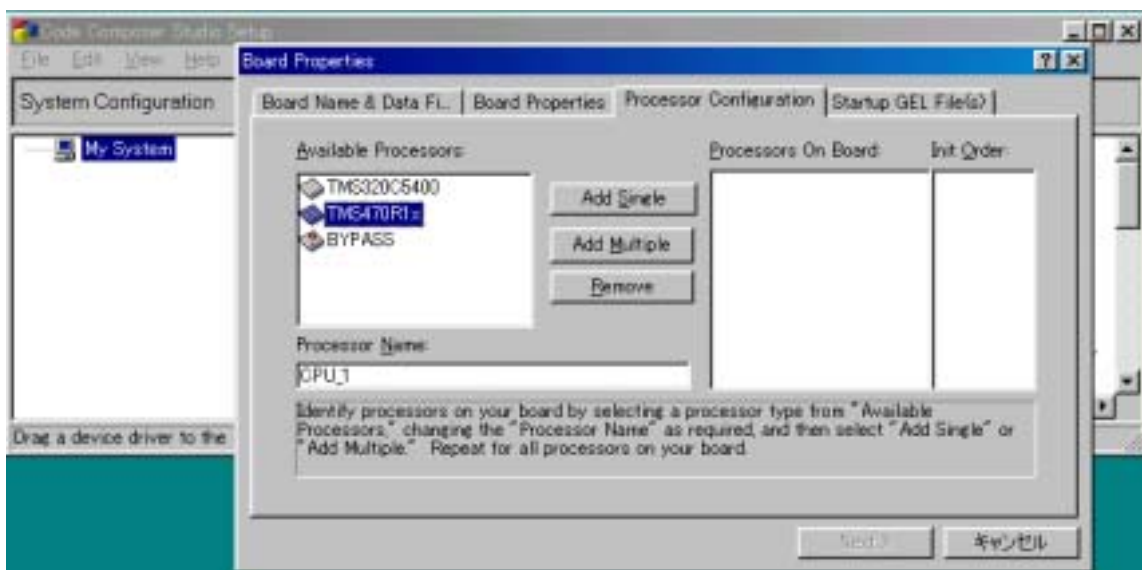


Click Next> button here. Then, next screen opens.



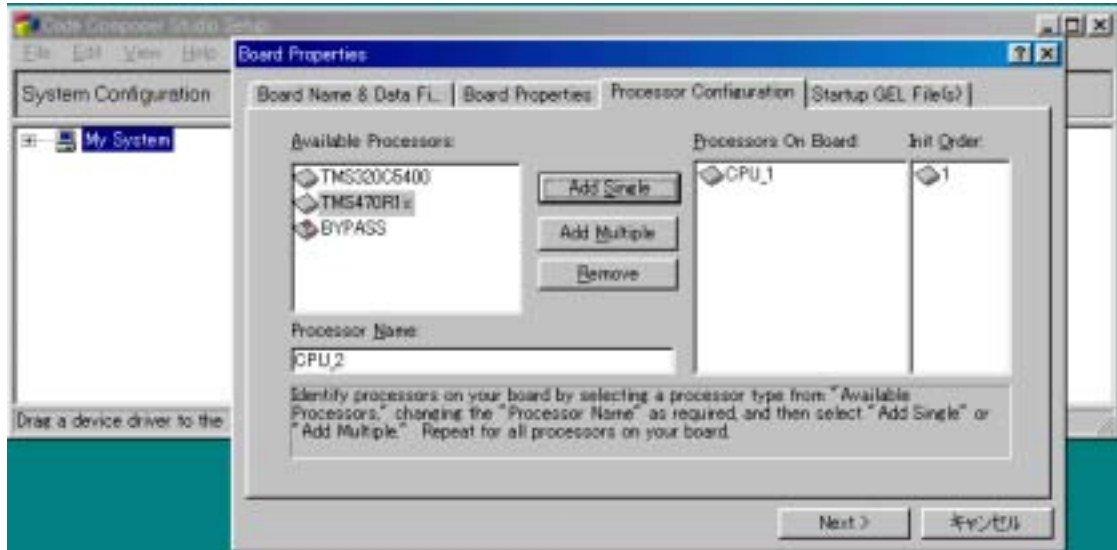
In case that I/O address is 0x240, click Next> button.

Setting screen for ARM device and DSP device appears. Be sure to set ARM(TMS470R1x) first.

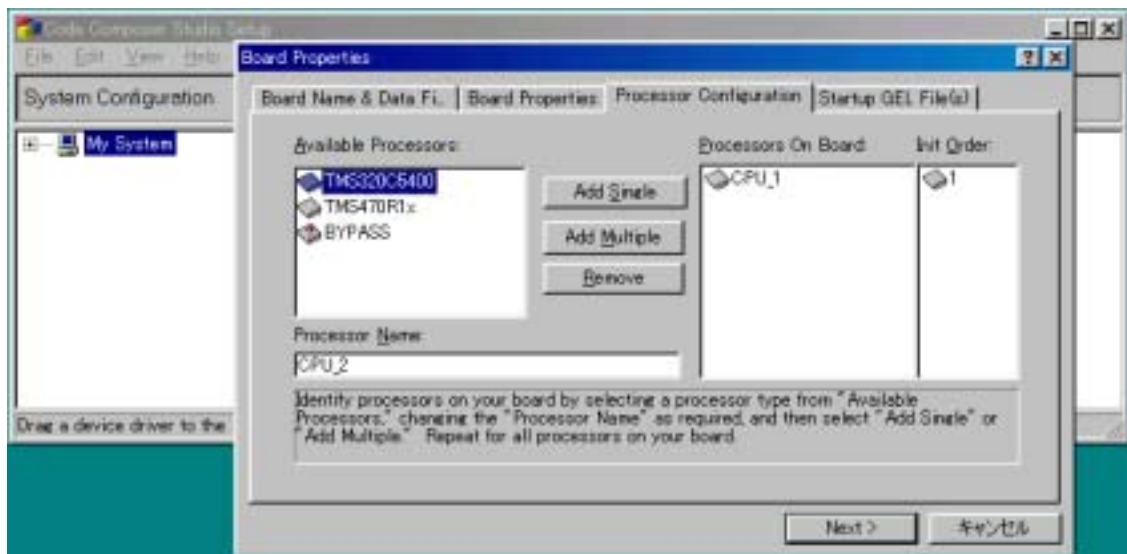


Select TMS470R1x in “Available Processors” window, and press “Add Single” button.

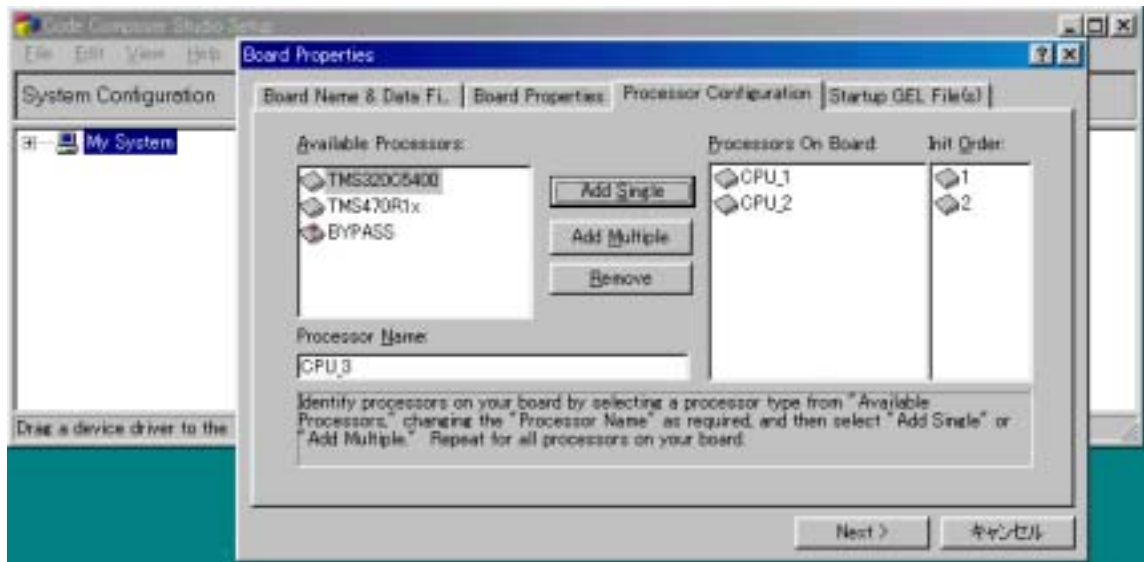
As CPU_1, setting as below is made in “Processors On Board” window which shows the selected processor.



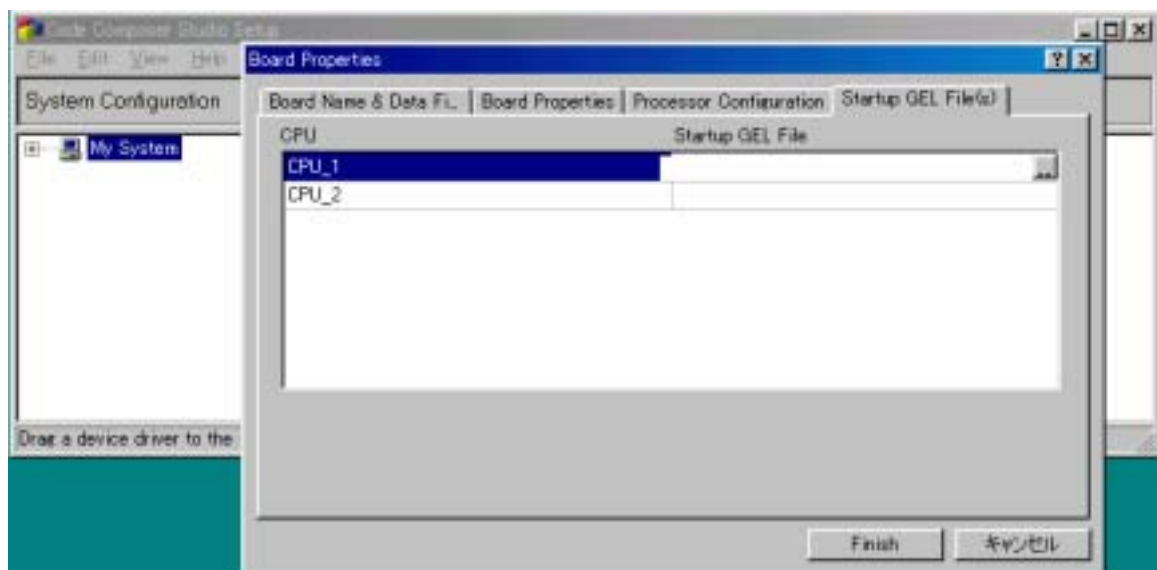
For setting up DSP (TMS320C5400), select TMS320C5400 in “Available Processors” window and press “Add Single” button.



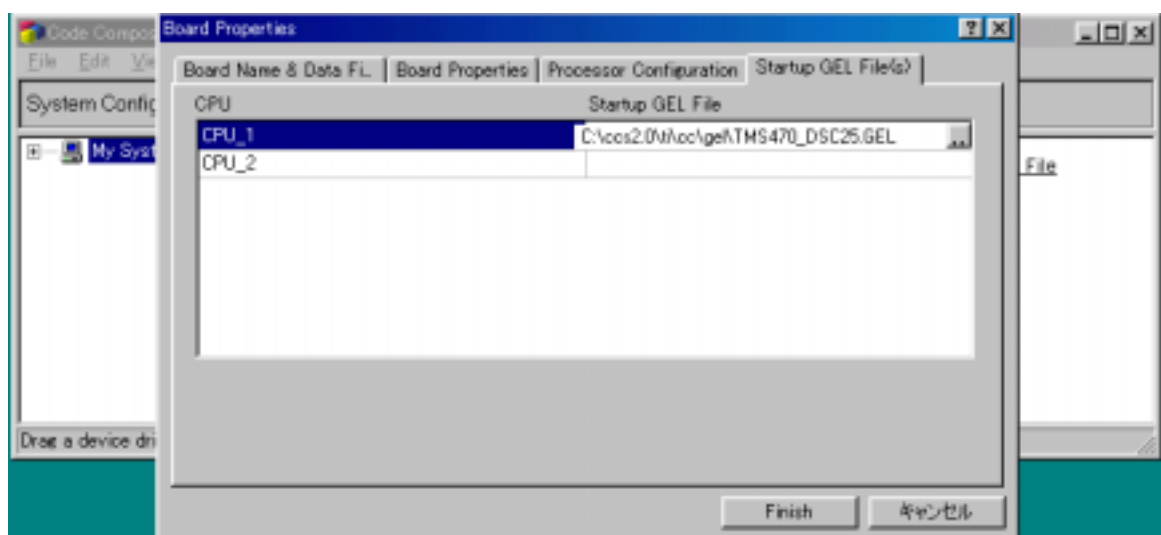
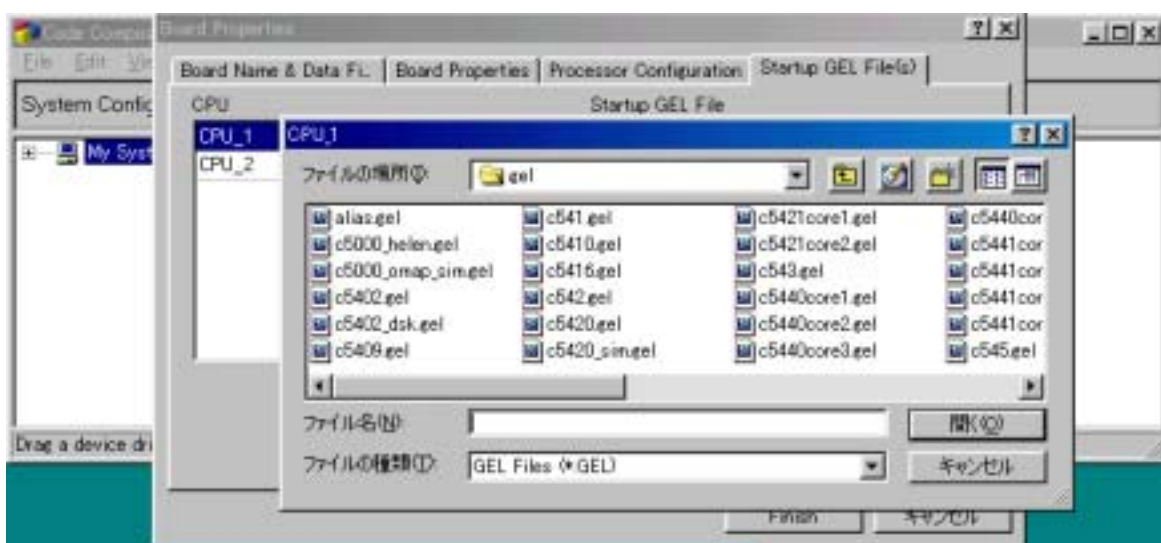
Setting of CPU_1 and CPU_2 are completed and shown in “Processors On Board” window as below.



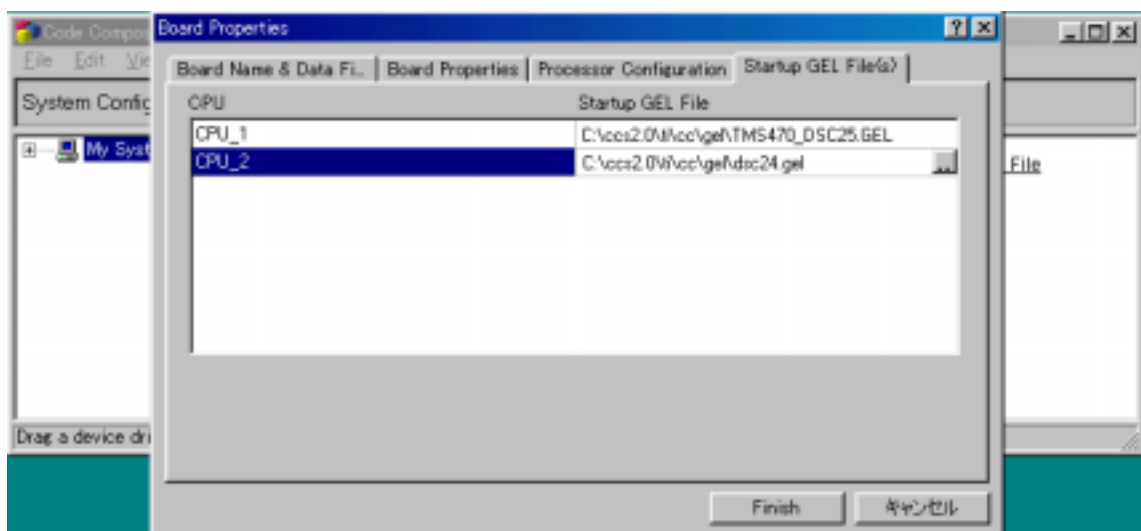
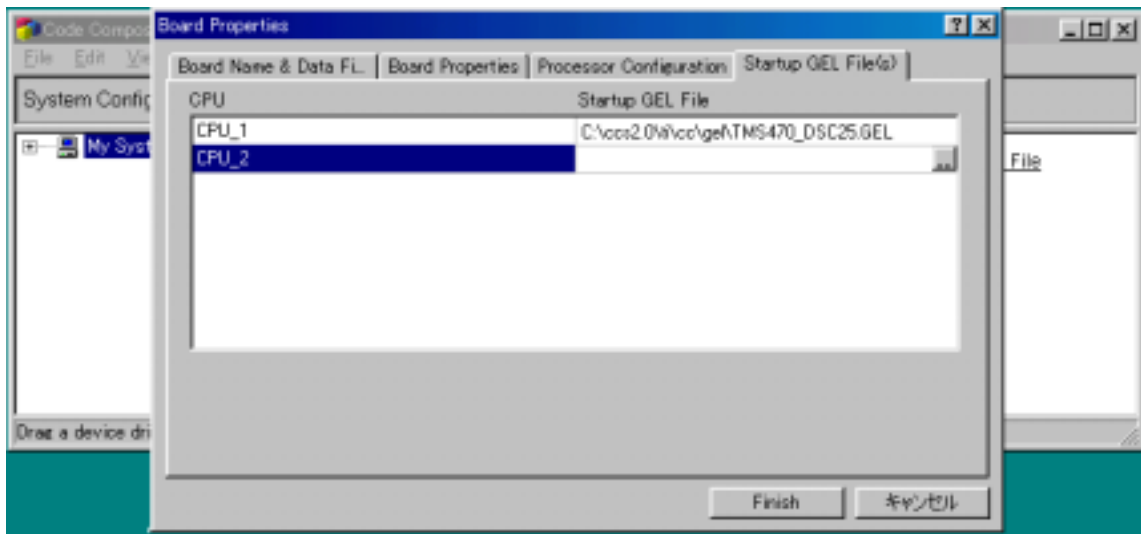
Press Next button. The location of gel file is asked.



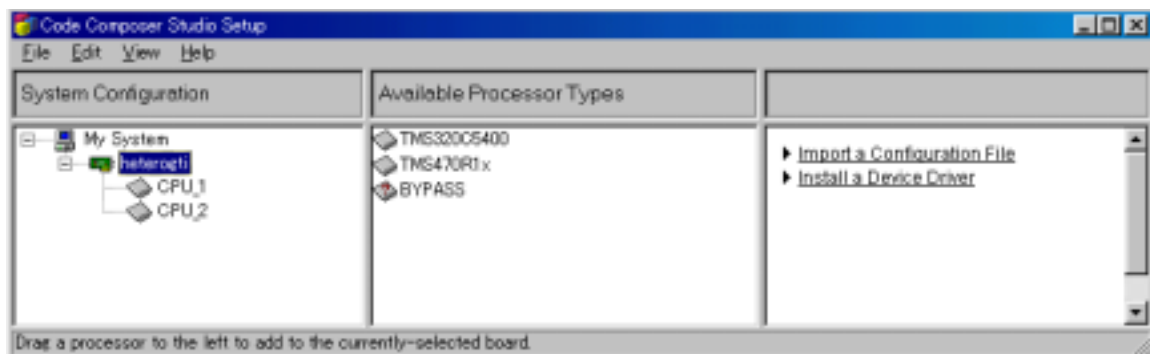
Press  button and specify the location of gel file, and then press “open” (O).



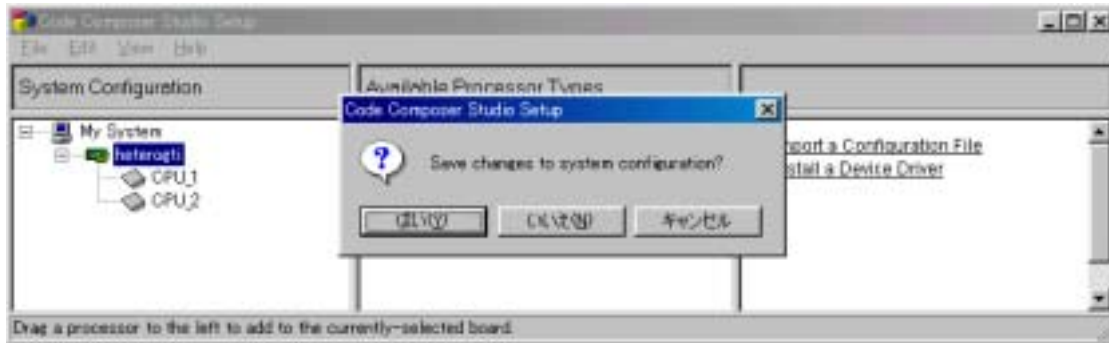
Configure CPU_2 in a similar way as CPU_1.



Press “Finish” button.



You will be asked “Save Changes to system configuration?” Be sure to press Yes (Y) and save this configuration.



Notice: Once pressed No (N), you must start these settings from the beginning.

Configuration of heterogeneous is completed.

After setting-up completed, you are asked whether you start up CCS on exit or not.



Press No (N) and complete the setting-up.

3-2. Powering of board

The power supply voltage on board subjects to be set at a stipulated voltage.

Connect all cables with board.

At this time, DC 5V power supply is still in OFF status.

Do not insert / remove the cable in ON status.

First, confirm the power supply switch, SW5 on board is OFF (turning down to front side)

- Insert XDS510 cable to JP20 JTAG.

- Insert power cable to J31.

Confirm that power cable RED connects to +5V and Black connects to GND adequately.

- Confirm that setting of jumper is consistent with setting of page.

- Emulation mode switches, SW3 and SW4

Confirm whether configurations of both TEST1 and TEST0 are “L” (front side) or not.

- DM270 operation mode setting switch, S2

Confirm that configuration of this switch is as Page 16.

- Device **TMS320DM270GHK**

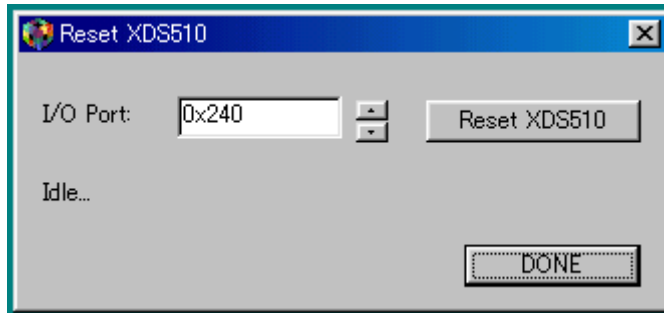
Confirm whether this is in U9 socket or not.

After confirmation above, apply DC 5V power supply and then rotate power supply switch, SW8 away (to far side).

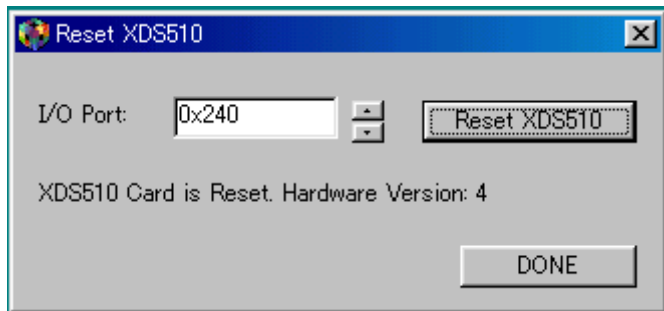
3-3. Resetting of XDS510

Reset XDS510.

PC: Press Start -> Code Composer Studio TMS470 -> RESETXDS510 and start program below.



Press RESET XDS510 button.



After reset, "XDS510 Card is Reset. Hardware Version: 4" is displayed as above.

Confirm this display and press DONE button.

3-4. Starting-up of CCS

After execution of RESETXDS510, start up CCS.

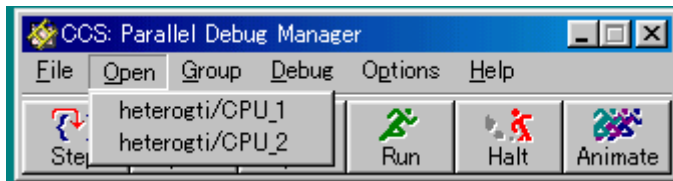
PC: To start up CCS, Press Start -> Program -> Texas Instruments -> Code Composer Studio 2 (OMAP) -> Setup Code Composer Studio.

Once started up successfully, a window below appears on PC display.

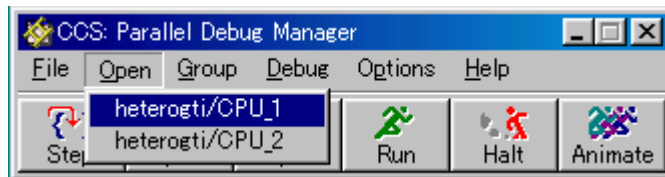


Starting up of ARM7 debugging window

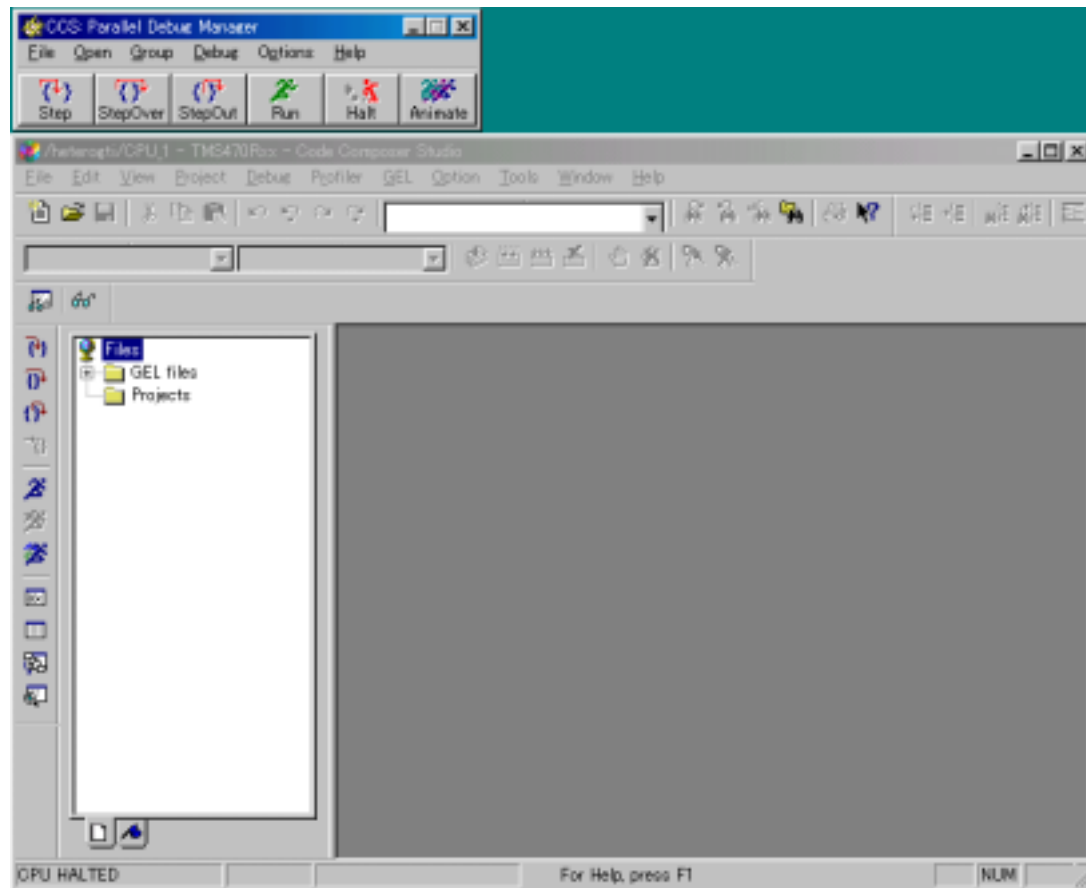
Click Open and a pop-up window of Parallel Debug Manager as below appears.



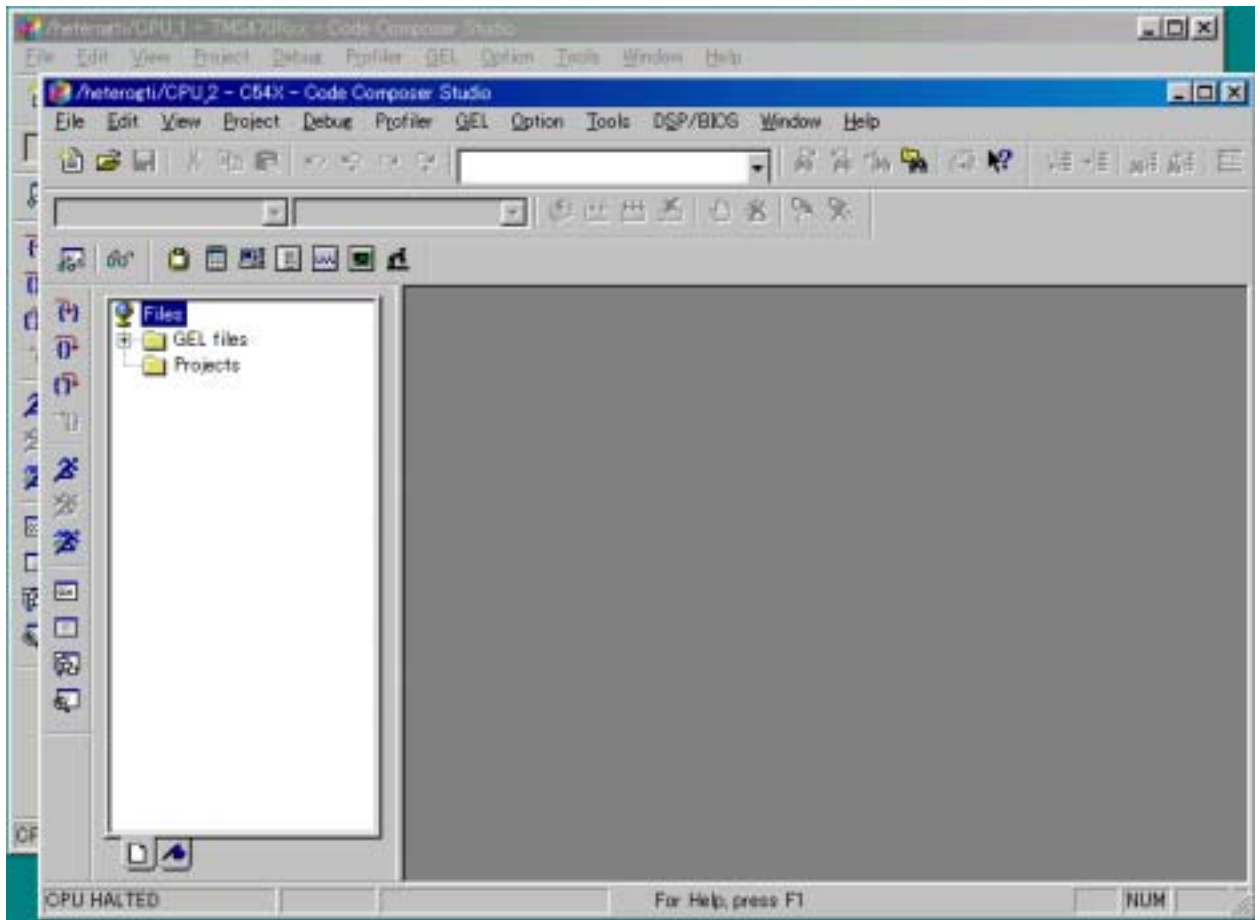
Click Heterogeneous Multi-Target (Texas Instruments)/CPU_1 by left button.



CPU debugger window of CPU_1 appears as below on display.



Press Parallel Debug Manager -> Open and select CPU_2. CPU_2 debugging window for DSP is displayed.



Now, start an operation using debugging window CPU_1 for ARM7.

Creating short-cut

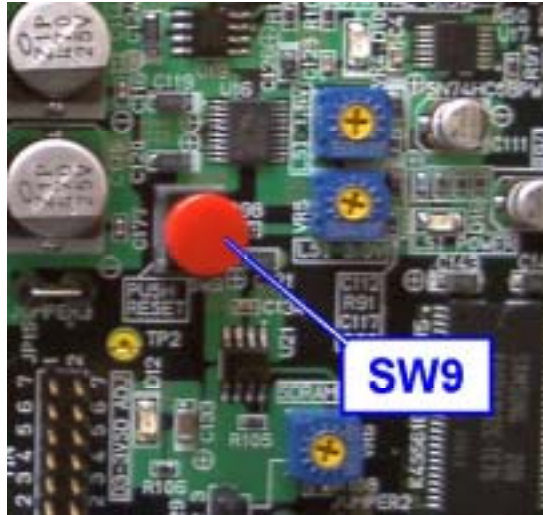
Create a short-cut icon for CCS470 on desktop.



3-5. Hard-reset of board

The hard-reset of board is done by pushing SW9 switch.

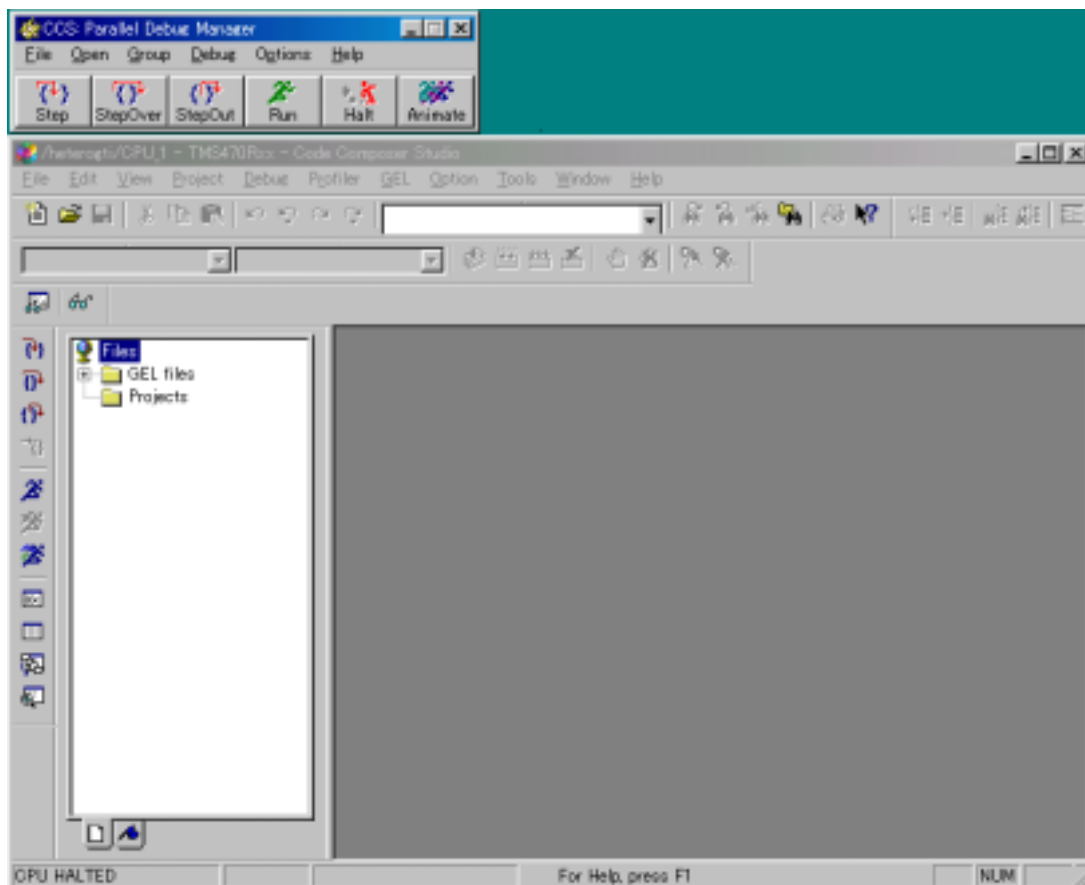
Now, reset the device on board by pushing SW9.



3-6. Starting-up of ARM debugger

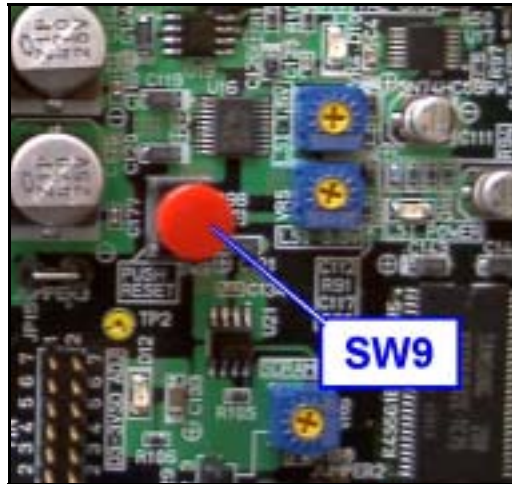
Click the short-cut icon for ARM7 CCS created already.

Once a pop-up window of Parallel Debug Manager appeared, click CPU_1 and start up the debugger for ARM7.



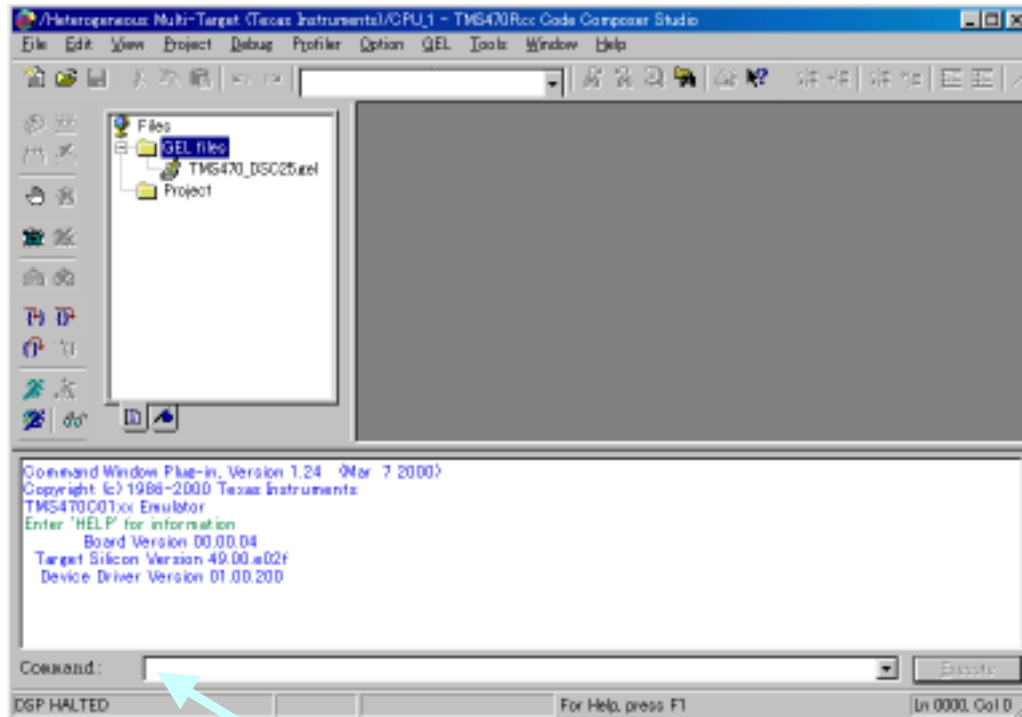
3-7. Hard-reset of board

Push SW9 once more and reset the device on board.



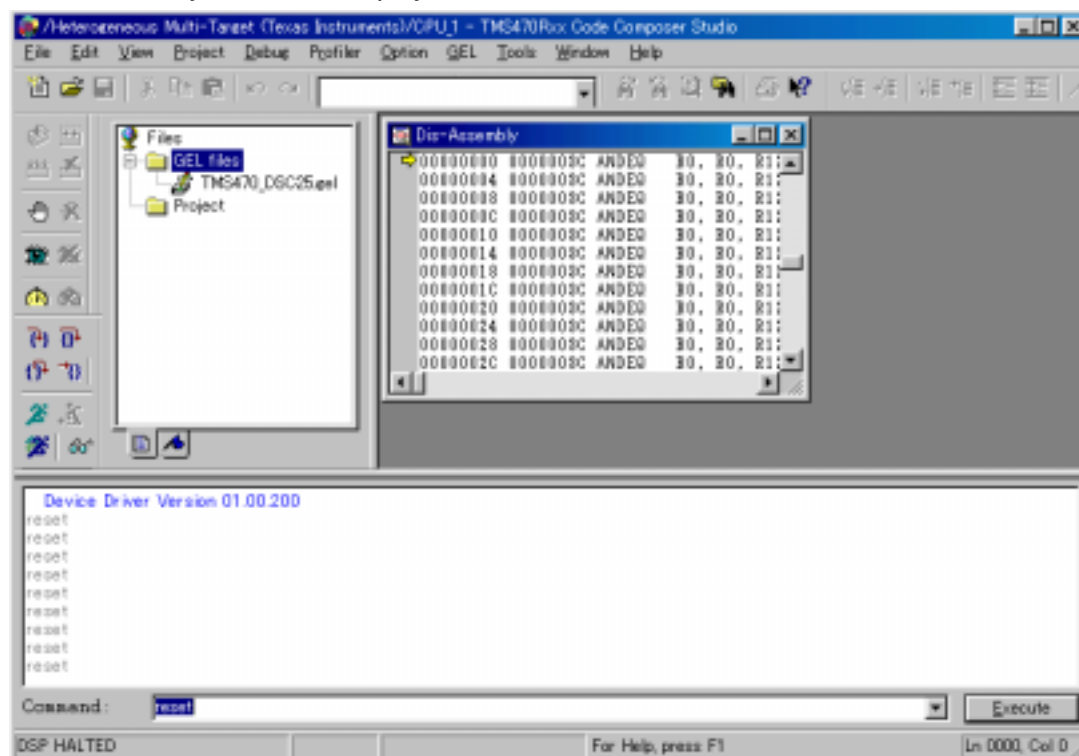
3-8. Resetting of CCS

Click Tools -> Command Window and open command window that is displayed at the bottom as below.



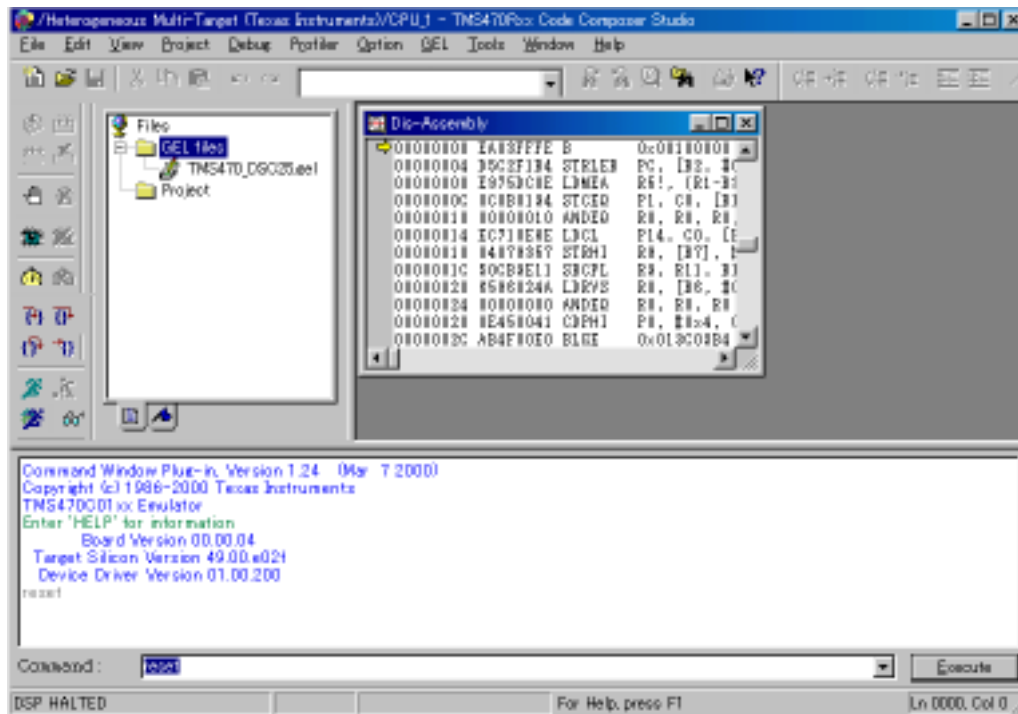
Enter **reset** in command window.

A Dis-Assembly window is displayed as below.



3-9. Display when resetting synchronization of CCS and device was done successfully

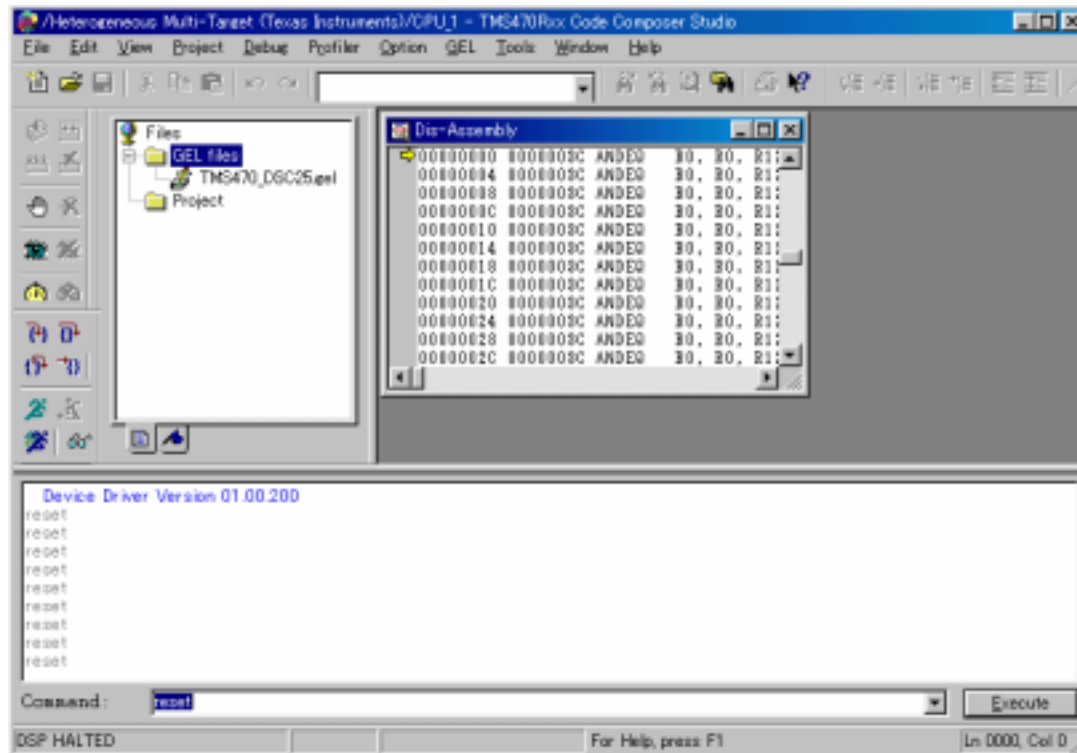
When starting up of SW9 which is push-button switch to hard-reset EVM board and CCS is well done, you can see B 0x100000 address order at 0x00000000 address of dis-assembly window.



Example for CCS display when resetting synchronization is normal

3-10. Display when resetting synchronization of CCS and device was not done successfully

It may be happened that you can not see B 0x100000 address at 0 address in dis-assembly window after setting-up.



This phenomenon is caused at emulation when device resetting and CCS resetting are not synchronous. It is not attributable to the board.

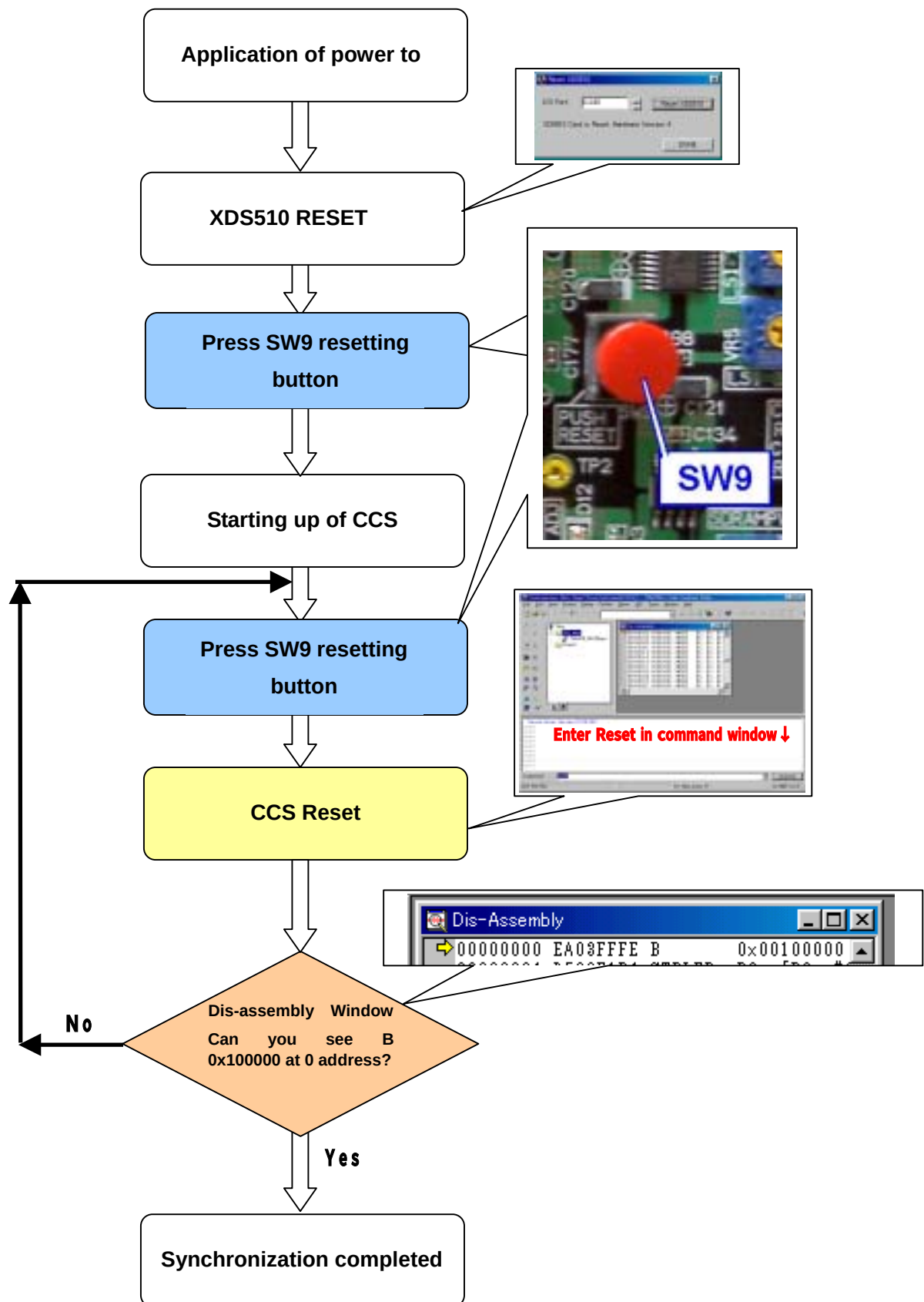
Under the condition of non-synchronization like this, normal operation is not expected in download of program, updating of memory and so on.

For a reliable synchronization of resetting of DM270 device and resetting of CCS, Reset XEVM-DM270GHK board by switching SW9 and CCS alternately till you see B 0x100000 at 0 address of program memory.

The synchronization flow of device resetting and CCS resetting is as follows.

In case that the synchronization is not made, start from hard resetting of board on Page. 32

3-11. Synchronization flow of device resetting and CCS resetting



3-12. Items to be confirmed when resettings are non-synchronous

In case that device resetting and CCS resetting are not synchronous even if it is tried several times, confirm the items below.

- Is device TMS320DM270GHK in U9 socket adequately?
-> Open U9 socket and confirm of presence or absence of device and insertion direction.

- Are configuration of CCS and emulation mode configuration of device consistent?
-> Confirm whether configuration in CCS Set-up and configuration of SW3 and SW4 are consistent or not.

- Is setting of operation mode dip switch S2 correct?
-> See Page.16 and confirm the correct setting.

- Is device clock provided adequately by U11 or Q2?
→Confirm that 27MHz oscillator module and 27MHz crystal resonator are implemented on U11 and Q2 respectively.
When operation is not made regardless of implementation above, confirm its oscillation.